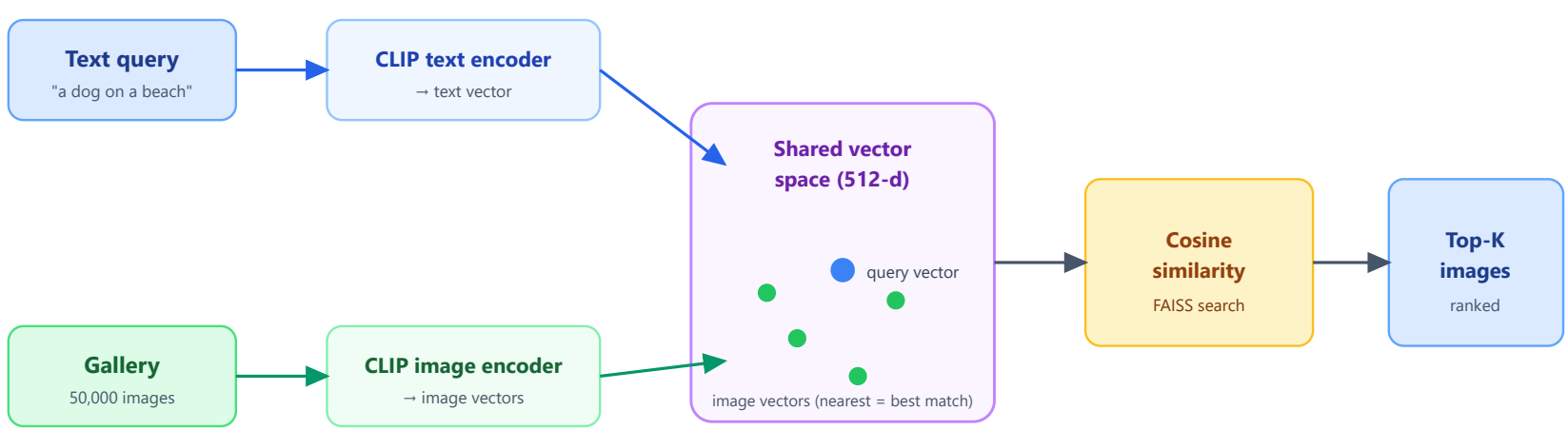
Text-to-Image Retrieval

System overview — how a text query finds the most relevant images



- **CLIP** is trained on image–caption pairs so a caption and its matching image end up **close together** in one shared space.
- **Offline (once):** each gallery image is encoded to a 512-d vector and stored in a **FAISS index**.
- **Online (per query):** the text query is encoded to a vector; the most similar image vectors (by **cosine similarity**) become the ranked results.
- Optional refinements: **BM25 keyword fusion** (hybrid) and **reranking** improve ordering.

Full engineering architecture: see the appendix page / docs/system-schematic.svg.

System Architecture (detail)

Offline indexing pipeline and online retrieval service

